

Project Name: ComicVerseAI
Team Members: P V Poornima, R Sandhiya, S Prathistha, Kiruthika K

PRODUCT BACKLOG, SPRINT SCHEDULE, AND ESTIMATION

Product backlog detailing functional requirements, user stories, estimations, priorities, assigned team members, and sprint delivery schedules for ComicVerseAI.

SPRINT	FUNCTIONAL REQUIREMENT (EPIC)	USER STORY NUMBER	USER STORY / TASK	STORY POINTS	PRIORITY	TEAM MEMBERS
Sprint-1	AI Script Breakdown	USN-1	As a user, I can enter a text story prompt so that Gemini LLM parses it into structured panel scripts.	2	High	P V Poornima
Sprint-1	AI Script Breakdown	USN-2	As a user, I receive panel JSON defining scene background, character prompts, and dialogue bubbles.	1	High	R Sandhiya
Sprint-1	Artwork Generation	USN-3	As a user, I can render comic artwork automatically using Stable Diffusion API matching selected art styles.	2	High	S Prathistha
Sprint-1	Interactive Canvas	USN-4	As a user, I can edit dialogues and speech bubble locations directly on an interactive canvas layout.	2	Medium	Kiruthika K
Sprint-2	PDF Export	USN-5	As a user, I can export the generated comic book into high-resolution, print-ready PDF formats.	1	High	Team

PLANNING SUMMARY & ALLOCATION

Sprint 1 (4 User Stories): Focuses on story generation via LLM, Stable Diffusion panel rendering, and canvas assembly UI.
Sprint 2 (1 User Story): Focuses on exporting multi-panel comic pages to PDF format using WeasyPrint compilation.

This document evaluates the quality of the **ComicVerseAI** codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

CODE LAYOUT CHECKLIST

S.NO	CODE QUALITY PARAMETER	DESCRIPTION	FOLLOWED	REMARKS
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space Python / ESLint standard
2	Proper File Structure	Files and folders are logically organized	Yes	Modular frontend / backend split
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive camelCase & snake_case
4	Function/Method Names	Functions are descriptively named	Yes	Action-oriented (e.g., generatePanel)
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings on key API routes
6	Modular Design	Code is split into reusable functions/modules	Yes	Decoupled AI services & UI views
7	No Redundant Code	Duplicate or unused code is removed	Yes	Clean refactored codebase
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-catch blocks on external APIs

REUSABLE COMPONENTS / MODULES

S.NO	COMPONENT / MODULE NAME	LANGUAGE / TECHNOLOGY	WHERE REUSED	REUSABILITY LEVEL
1	Gemini Prompt Parser	Python / FastAPI	Script breakdown & dialogue generator	High
2	Stable Diffusion Wrapper	Python / Diffusers	All panel background & character rendering	High
3	Comic Canvas Grid Component	React / TypeScript	Interactive editor & page layout view	High
4	Speech Bubble Renderer	CSS3 / SVG	Panel dialogue overlays	Medium
5	PDF Export Engine	Python / WeasyPrint	Final comic document download engine	High

OVERALL CODE QUALITY ASSESSMENT

ASPECT	RATING (1-5)	COMMENTS
Code Layout & Structure	5 / 5	Clean separation between frontend UI components and backend FastAPI endpoints.
Readability	4.8 / 5	Consistent identifier naming and clear variable scoping throughout modules.
Reusability	4.9 / 5	Modular AI pipeline wrappers enable easy swapping or expansion of image generation engines.
Documentation / Comments	4.7 / 5	Comprehensive API endpoint documentation and inline comments explaining prompt parsers.
Overall Score	4.85 / 5	High production quality code meeting all clean architecture criteria.

CODING & SOLUTION

This section evaluates the quality and completeness of the implemented **ComicVerseAI** code and overall solution. The submission demonstrates working AI comic creation, clean code practices, and alignment with functional requirements.

SOLUTION INSTRUCTIONS & REPOSITORY

- Code is structured modularly with clear docstrings and typing across frontend and backend tiers.
- The solution parses natural language stories, constructs structured panel JSON, synthesizes AI art, overlays speech bubbles, and compiles output as PDF.

SOLUTION SUMMARY TABLE

FIELD	DETAILS
Repository Link / URL	https://github.com/ComicVerseAI/comicverseai-core
Programming Language(s)	Python 3.11, TypeScript, JavaScript (ES6+)
Framework(s) Used	FastAPI (Backend), React.js, Tailwind CSS, Diffusers API, WeasyPrint
Key Features Implemented	Prompt Breakdown Engine, Image Synthesis Pipeline, Dynamic Speech Bubbles, Printable PDF Export
Pending / Incomplete Features	Multi-character persistent voice synthesis (Planned for v2.0)
Setup / Run Instructions	<code>pip install -r requirements.txt && uvicorn main:app --reload (Backend)</code> <code>npm install && npm start (Frontend)</code>

CODE QUALITY CHECKLIST

S.NO	CRITERIA	STATUS (YES / NO)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments: All units tested via PyTest and Jest framework integration. Automated CI/CD pipelines verify static typing and syntax formatting prior to deployment.

NO. OF FUNCTIONAL FEATURES INCLUDED IN THE SOLUTION

This section lists all functional features included in the **ComicVerseAI** project solution, describing purpose, module association, and current implementation status.

S.NO	FEATURE NAME	FEATURE DESCRIPTION	MODULE / COMPONENT	STATUS
1	Story Prompt Parsing	Parses natural text into multi-panel comic storyboard JSON	AI Script Engine	Done
2	Panel Image Synthesis	Generates high-resolution artwork per panel description	Diffusers Service	Done
3	Speech Bubble Placement	Interactive drag-and-drop overlay for dialogue bubbles	Canvas UI	Done
4	Style Selector	Allows switching between Manga, Superhero, and Webtoon styles	Style Engine	Done
5	Printable PDF Exporter	Compiles all comic pages into print-ready PDF files	Export Engine	Done
6	User Auth & History	Allows users to save and edit previously created comics	Backend Auth	In Progress

FEATURE SUMMARY METRICS

METRIC	COUNT / VALUE
Total Features Planned	6 Features
Total Features Implemented	5 Completed / 1 In Progress
Core / Must-Have Features	5 Features
Additional / Nice-to-Have Features	1 Feature (User History)
Features Tested & Verified	5 Features (100% Core Pass Rate)

FEATURE CATEGORY BREAKDOWN

S.NO	CATEGORY	FEATURES IN CATEGORY	EXAMPLE FEATURES
1	User Interface (UI)	Interactive Canvas, Speech Bubble Editor	Drag-and-drop bubbles, inline text edit, layout preview
2	Backend / Logic	AI Script Breakdown, JSON Parser	Gemini LLM story prompt to panel JSON converter
3	Database / Storage	Panel Asset Cache, User Projects	PostgreSQL storage for panel JSON and generated image URLs
4	API / Integration	Stable Diffusion API, Gemini API	External AI model endpoint connectors and status polling
5	Security / Auth	JWT Authentication, API Key Guard	Secure token validation for user sessions and API access

Final Verification Sign-Off

This document verifies that the **ComicVerseAI** platform meets all initial architectural, code readability, reusability, and functional criteria outlined in the project planning sprint schedules.